Cell Biology

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Lecture:4

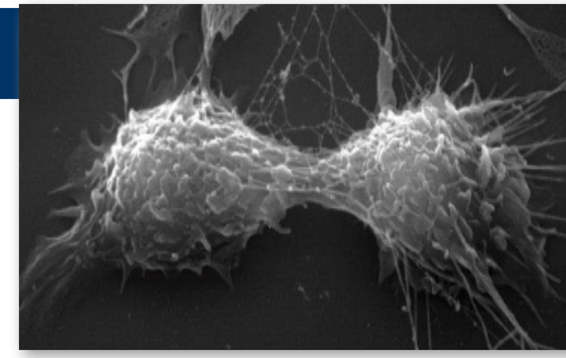
Frequency of cell division, Cell cycle

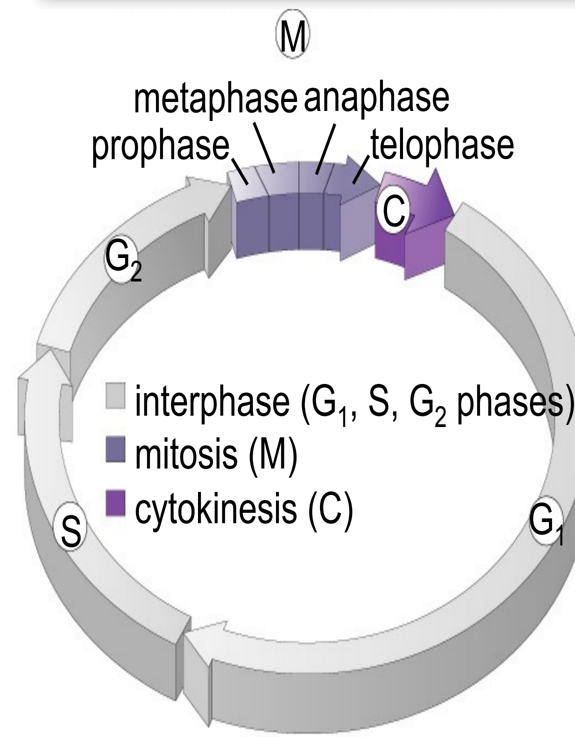
Introduction

- The ability of organisms to reproduce their kind is one characteristic that best distinguishes living things from nonliving matter.
- The continuity of life from one cell to another is based on the reproduction of cells via cell division.
- This division process occurs as part of the cell cycle, the life of a cell from its origin in the division of a parent cell until its own division into two.

Frequency of cell division

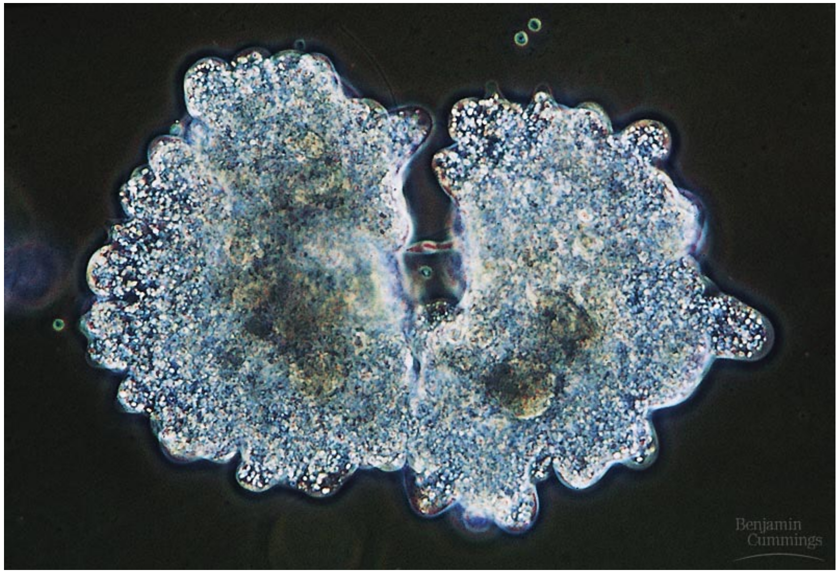
- Frequency of cell division varies by cell type.
  - embryo
    - cell cycle < 20 minute
  - skin cells
    - divide frequently throughout life
    - 12-24 hours cycle
  - liver cells
    - retain ability to divide, but keep it in reserve
    - divide once every year or two
  - mature nerve cells & muscle cells
    - do not divide at all after maturity
    - permanently in G_0





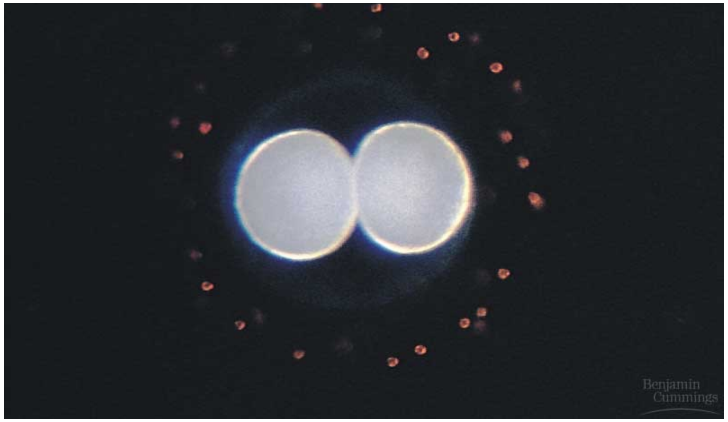
Cell division functions in reproduction, growth, and repair

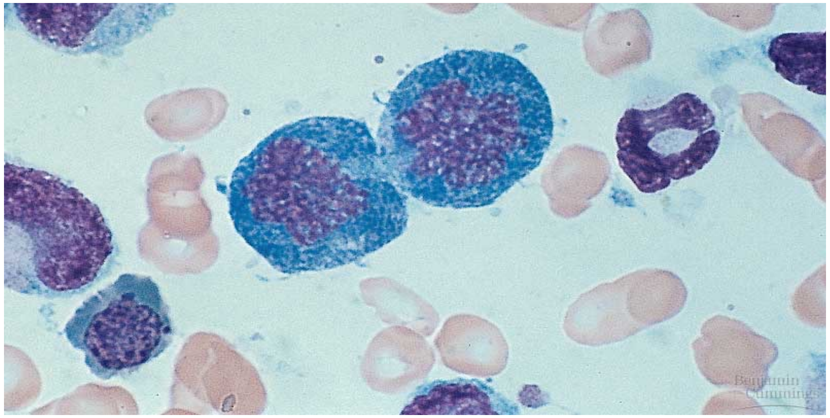
- The division of a unicellular organism reproduces an entire organism, increasing the population.
- Cell division on a larger scale can produce progeny for some multicellular organisms.
  - This includes organisms that can grow by cuttings or by fission.



Cell division functions in reproduction, growth, and repair

- Cell division is also central to the development of a multicellular organism that begins as a fertilized egg or zygote.



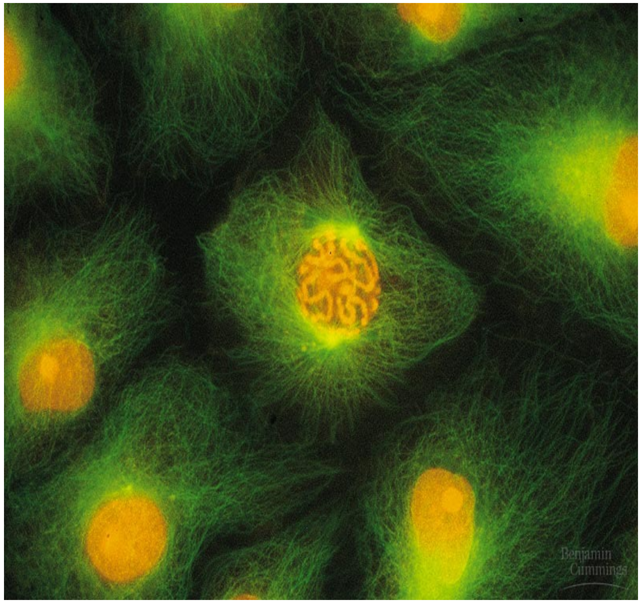


Cell division functions in reproduction, growth, and repair

- Cell division requires the distribution of identical genetic material - DNA - to two daughter cells.
  - What is remarkable is the fidelity with which DNA is passed along, without dilution, from one generation to the next.
- A dividing cell duplicates its DNA, allocates the two copies to opposite ends of the cell, and then splits into two daughter cells.

Cell division functions in reproduction, growth, and repair

- DNA molecules are packaged into **chromosomes**.
  - Every eukaryotic species has a characteristic number of chromosomes in the nucleus.
    - Human somatic cells (body cells) have 46 chromosomes.
    - Human gametes (sperm or eggs) have 23 chromosomes, half the number in a somatic cell.



Cell division functions in reproduction, growth, and repair

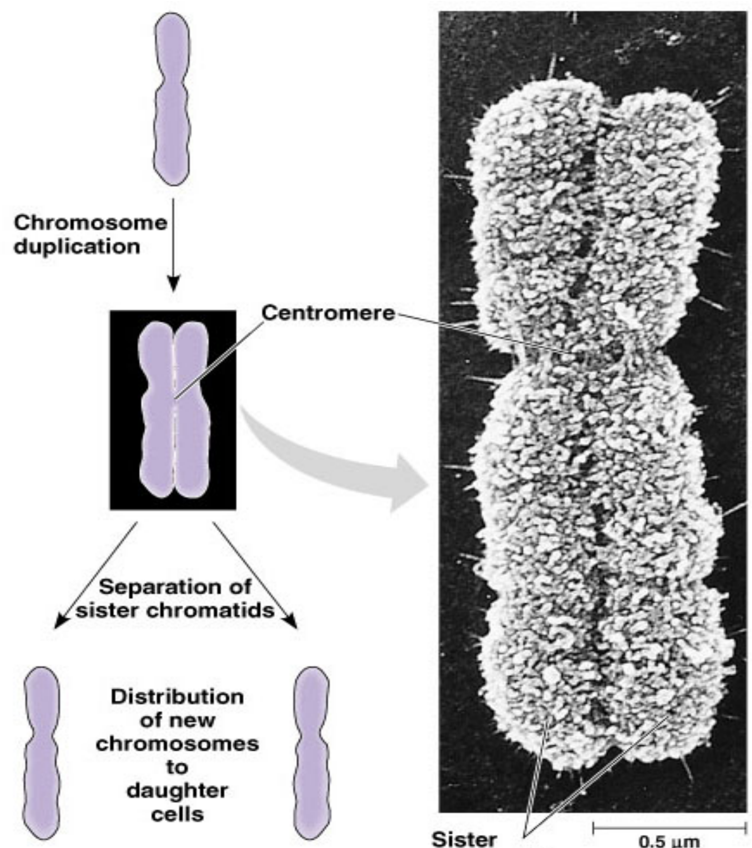
- Each eukaryotic chromosome consists of a long, linear DNA molecule.
- Each chromosome has hundreds or thousands of genes, the units that specify an organism’s inherited traits.

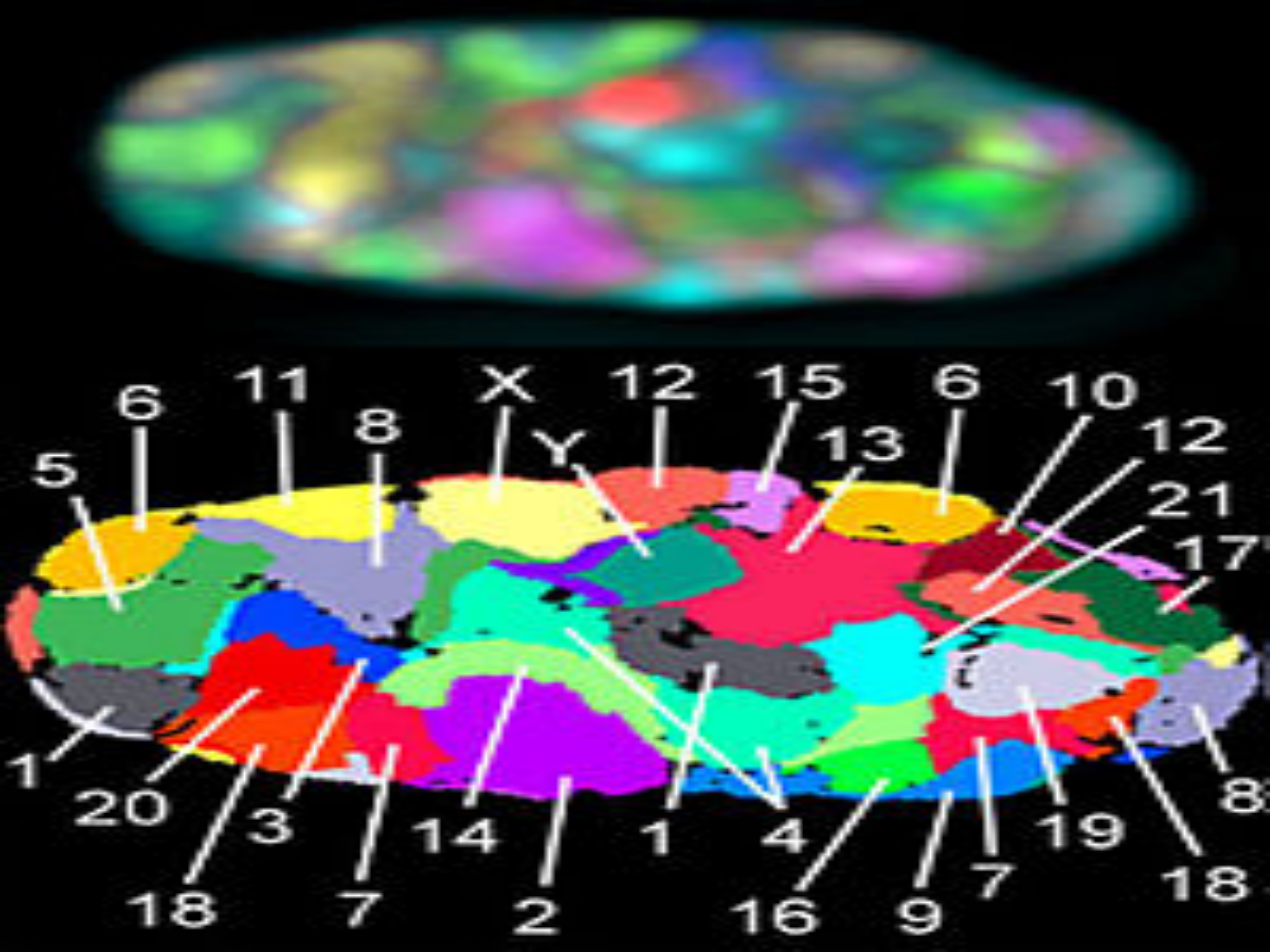
Cell division functions in reproduction, growth, and repair

- Associated with DNA are proteins that maintain its structure and help control gene activity.
- This DNA-protein complex, chromatin, is organized into a long thin fiber.
- After the DNA duplication, chromatin condenses, coiling and folding to make a smaller package

Cell division functions in reproduction, growth, and repair

- Each duplicated chromosome consists of two **sister chromatids** which contain identical copies of the chromosome's DNA.
- As they condense, the region where the strands connect shrinks to a narrow area, is the **centromere**.
- Later, the sister chromatids are pulled apart and repackaged into two new nuclei at opposite ends of the parent cell.





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X
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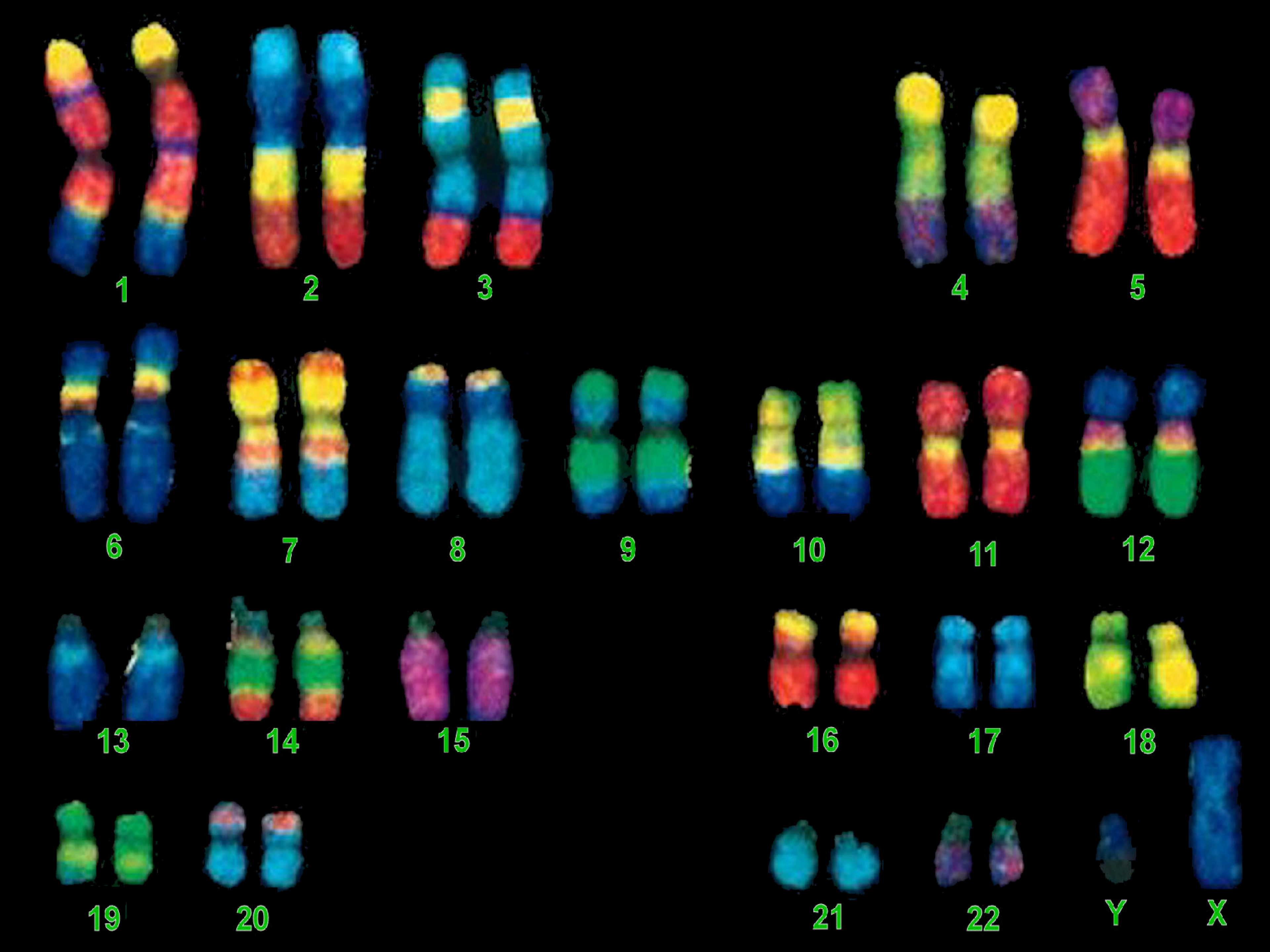
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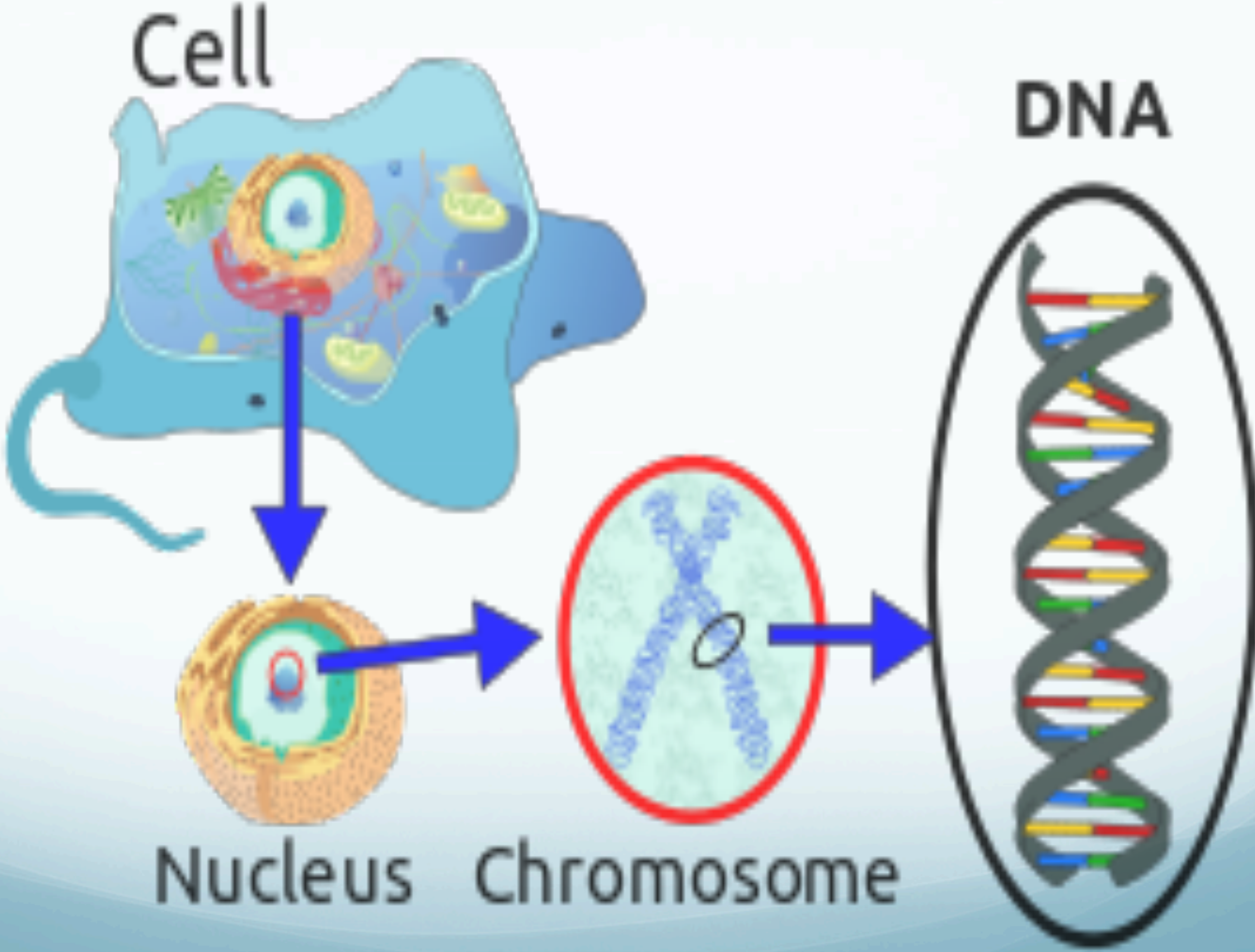
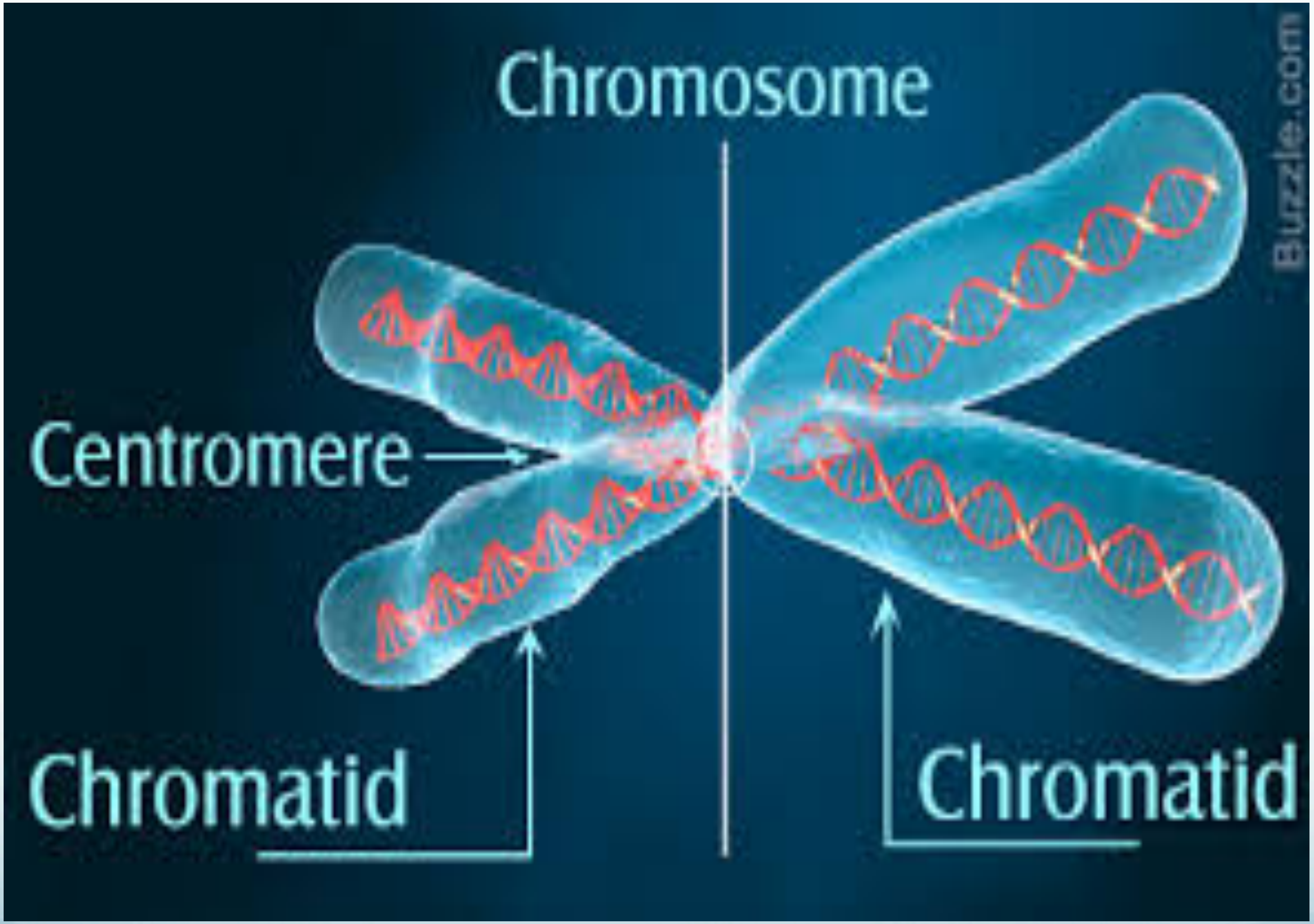
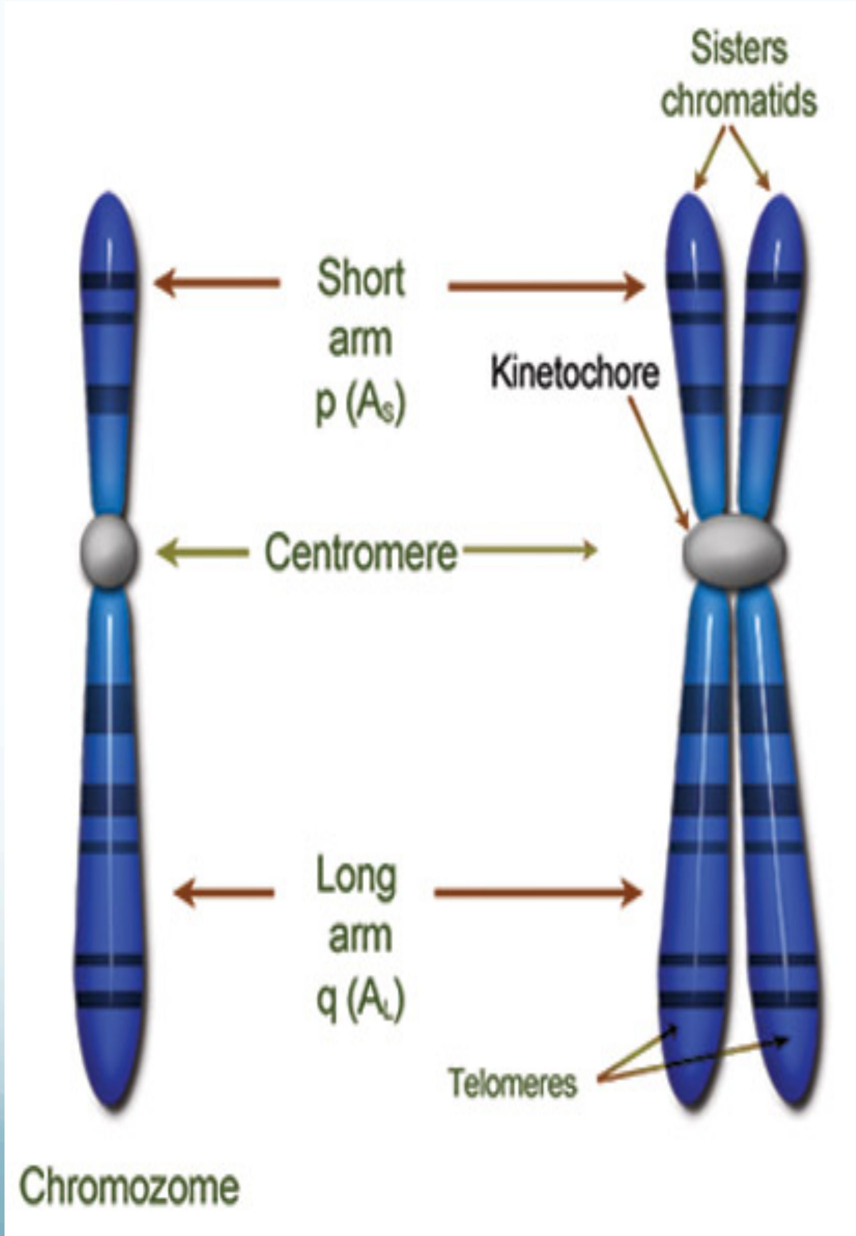
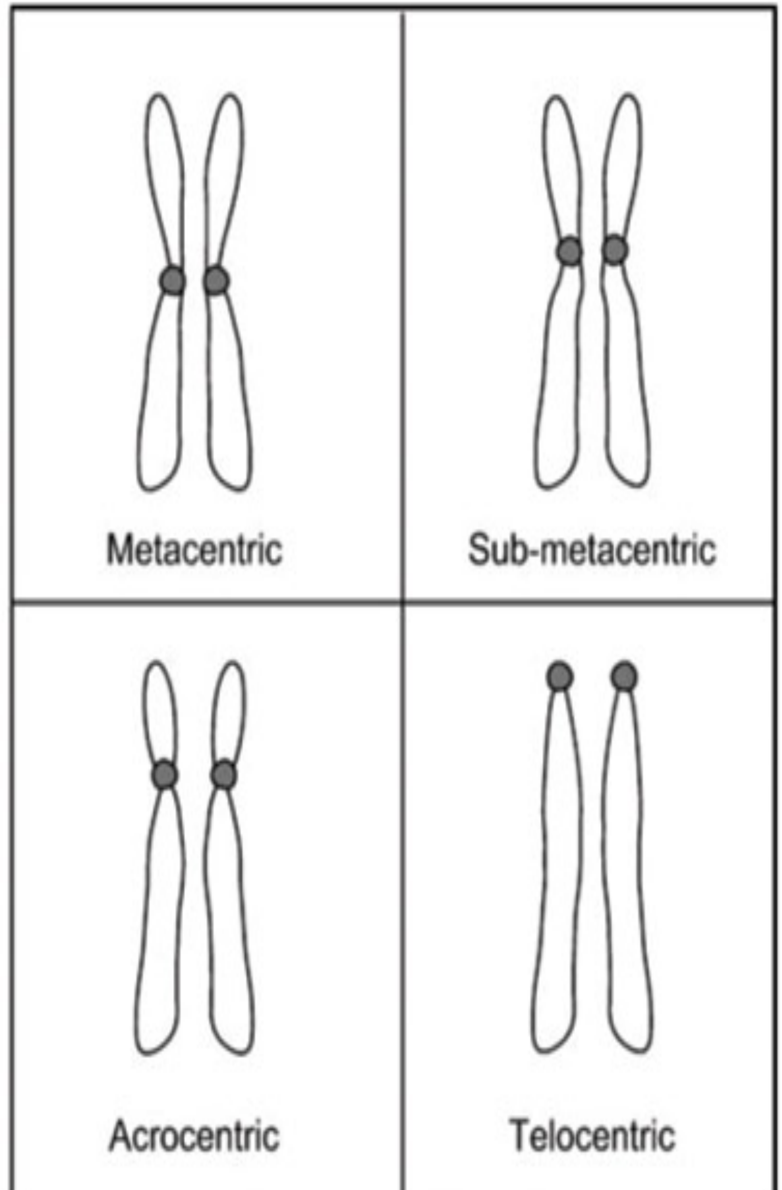
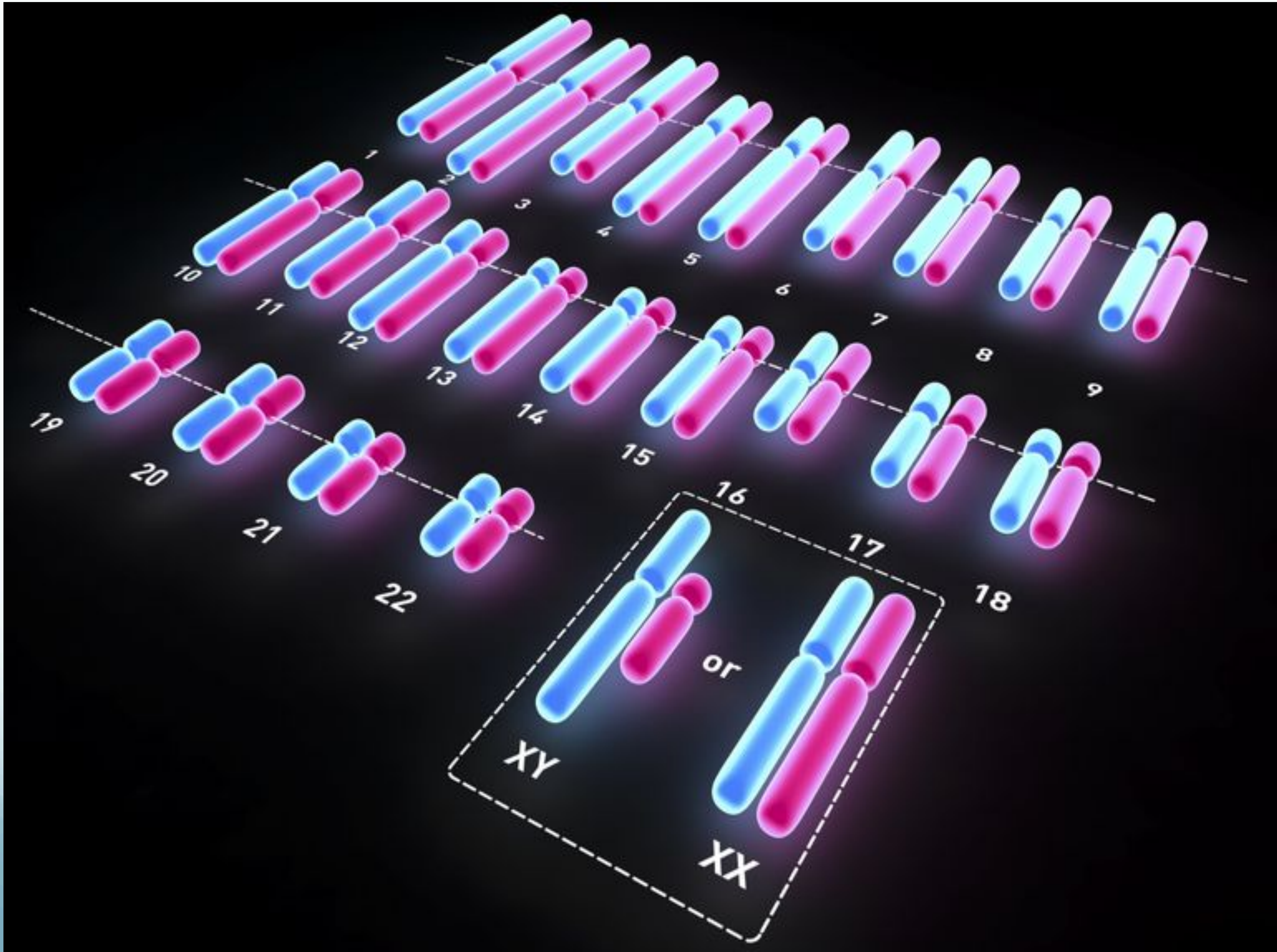

Cell
DNA
Nucleus
Chromosome


Chromosome
Centromere
Chromatid
Chromatid
Buzzle.com


Sisters
chromatids
Short
arm
p (A_S)
Kinetochore
Centromere
Long
arm
q (A_L)
Telomeres
Chromozome


Metacentric
Sub-metacentric
Acrocentric
Telocentric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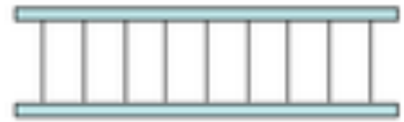
XY
or
XX

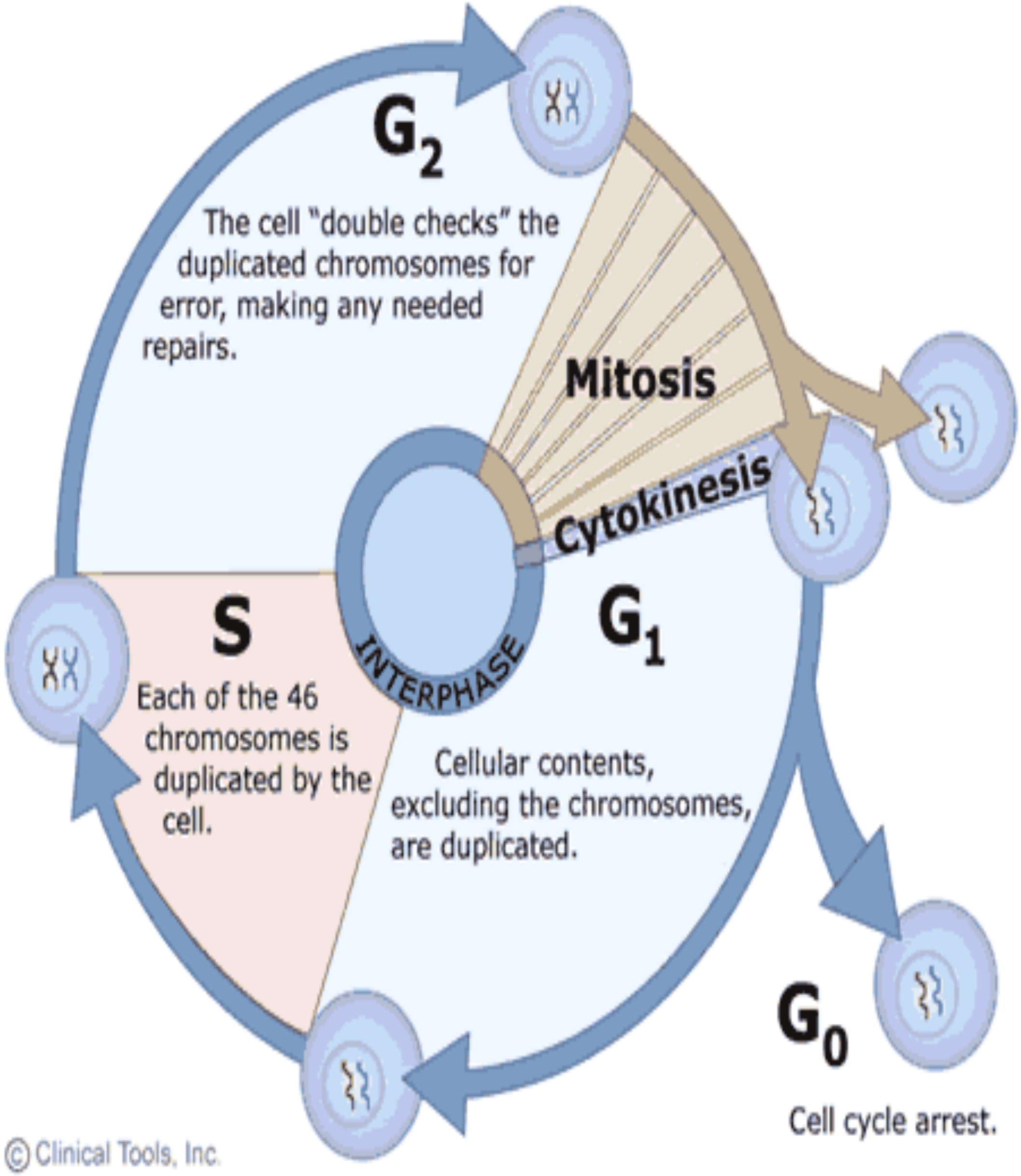
The chemical that carries the genetic code which controls how an organism develops	Chromosome		
A unit of inheritance giving instructions for a cell on how to make one type of protein	protein.		
Chemicals in living things made of a chain of amino acids. Have many jobs in living things	Gene		
Long thin threadlike structure in the nucleus of a cell. Made from DNA.	DNA		

Cell division functions in reproduction, growth, and repair

- The process of the formation of the two daughter nuclei, **mitosis**, is usually followed by division of the cytoplasm, **cytokinesis**.
- These processes take one cell and produce two cells that are the genetic equivalent of the parent.

The cell cycle

- Actively dividing eukaryote cells pass through a series of stages known collectively as the cell cycle. two gap phases (G1 and G2); an S (synthesis) phase, in which the genetic material is duplicated; and an M phase, in which mitosis partitions the genetic material and the cell divides


G2
The cell "double checks" the duplicated chromosomes for error, making any needed repairs.
Mitosis
Cytokinesis
S
Each of the 46 chromosomes is duplicated by the cell.
INTERPHASE
G1
Cellular contents, excluding the chromosomes, are duplicated.
G0
Cell cycle arrest.
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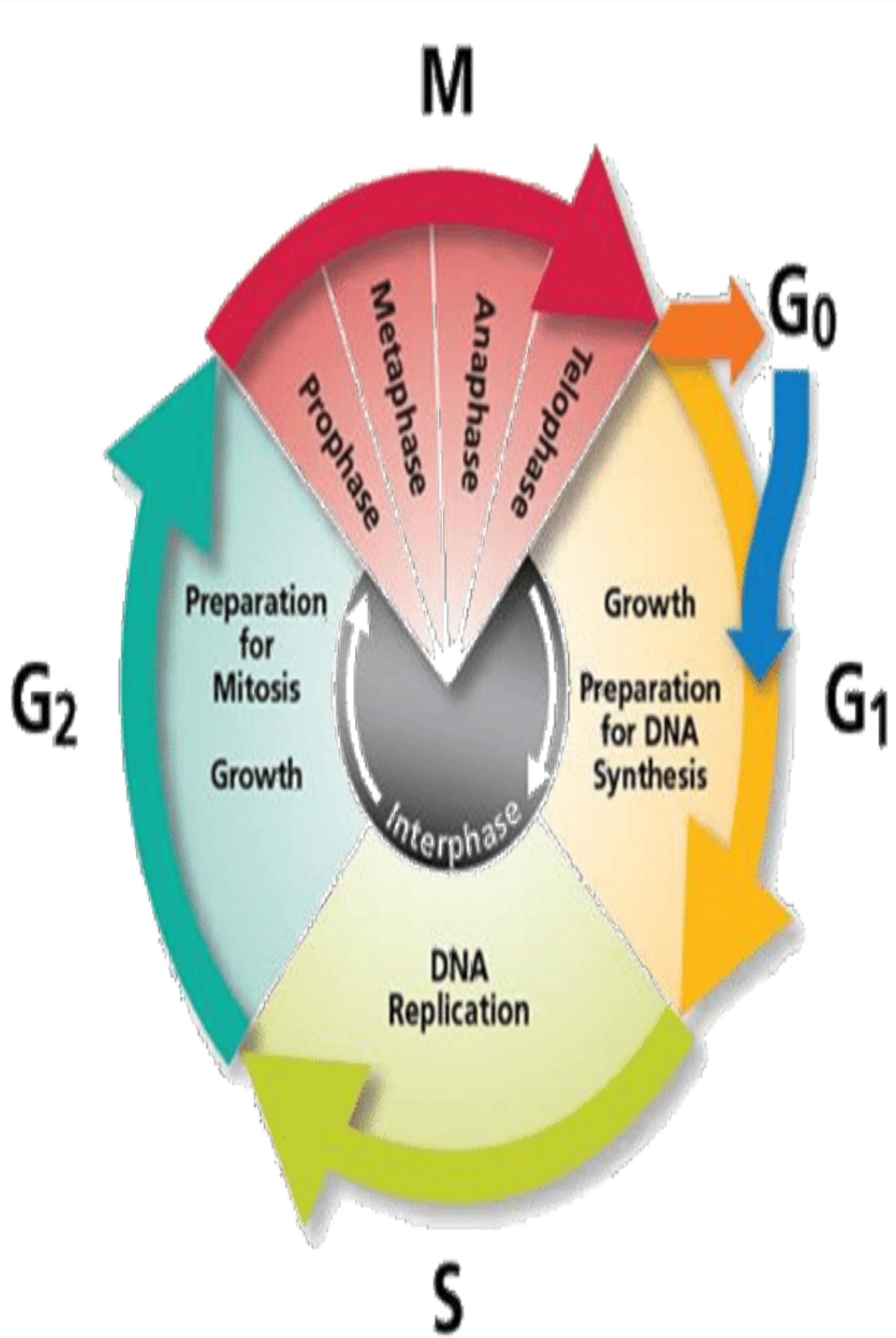
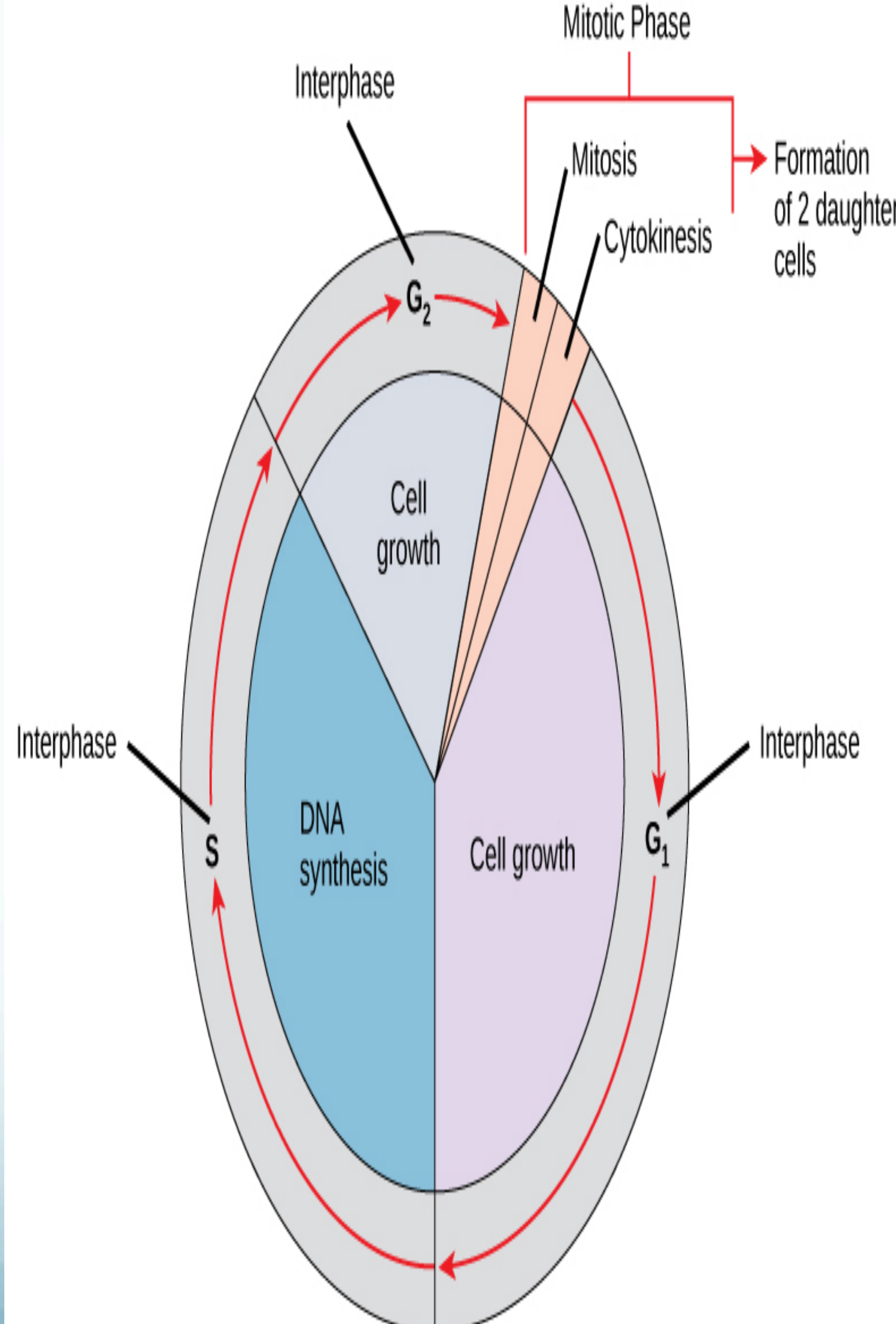

M
G_0
G_1
G_2
S
Prophase
Metaphase
Anaphase
Telophase
Preparation for Mitosis
Growth
Growth
Preparation for DNA Synthesis
DNA Replication
Interphase


Mitotic Phase
Interphase
Mitosis
Formation of 2 daughter cells
Cytokinesis
G_2
Cell growth
Interphase
DNA synthesis
Cell growth
Interphase
S
G_1

Cell cycle

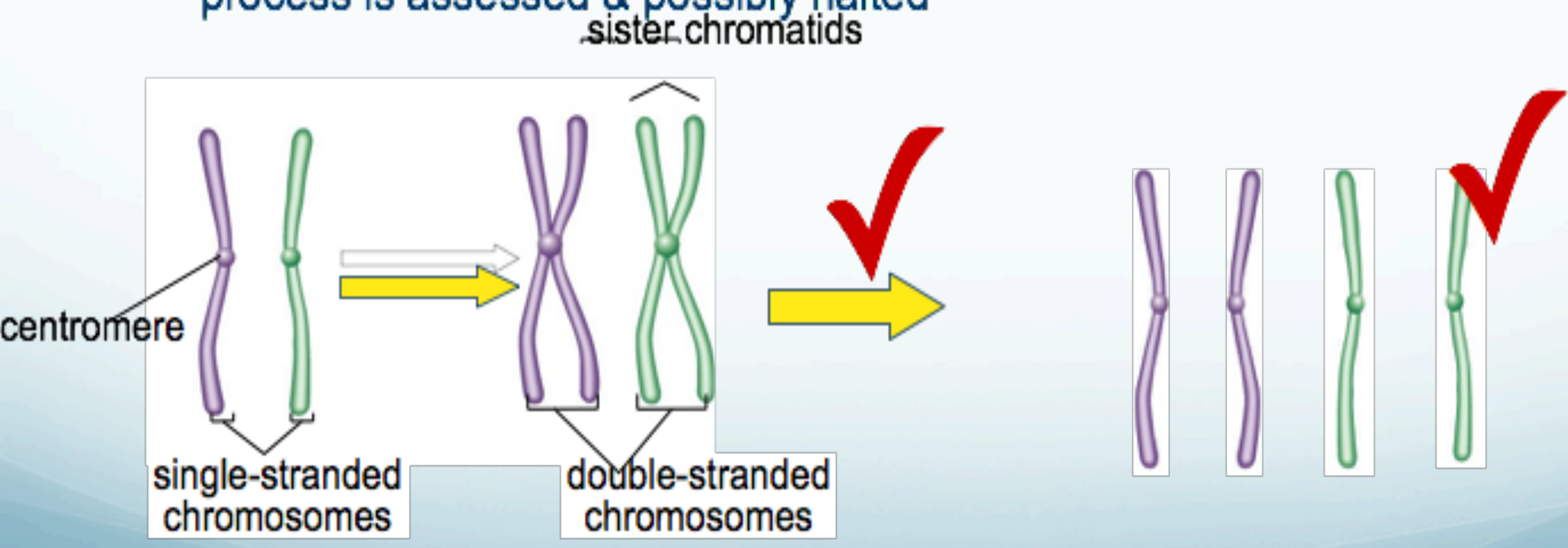
- **G1 Phase**
- The first stage of interphase is called the G1 phase, or first gap, because little change is visible. However, during the G1 stage, the cell is quite active at the biochemical level.
- The cell is accumulating the building blocks of chromosomal DNA and the associated proteins, as well as accumulating enough energy reserves to complete the task of replicating each chromosome in the nucleus.

- **S Phase**
- Throughout interphase, nuclear DNA remains in a semi-condensed chromatin configuration. In the S phase(synthesis phase), DNA replication results in the formation of two identical copies of each chromosome—sister chromatids—that are firmly attached at the centromere region. At this stage, each chromosome is made of two sister chromatids and is a duplicated chromosome. The centrosome is duplicated during the S phase

- **G2 Phase**
- In the G2 phase, or second gap, the cell replenishes its energy stores and synthesizes the proteins necessary for chromosome manipulation. Some cell organelles are duplicated, and the cytoskeleton is dismantled to provide resources for the mitotic spindle. There may be additional cell growth during G2. The final preparations for the mitotic phase must be completed before the cell is able to enter the first stage of mitosis.
- The period between mitotic divisions that is, G1, S and G2 - is known as interphase.

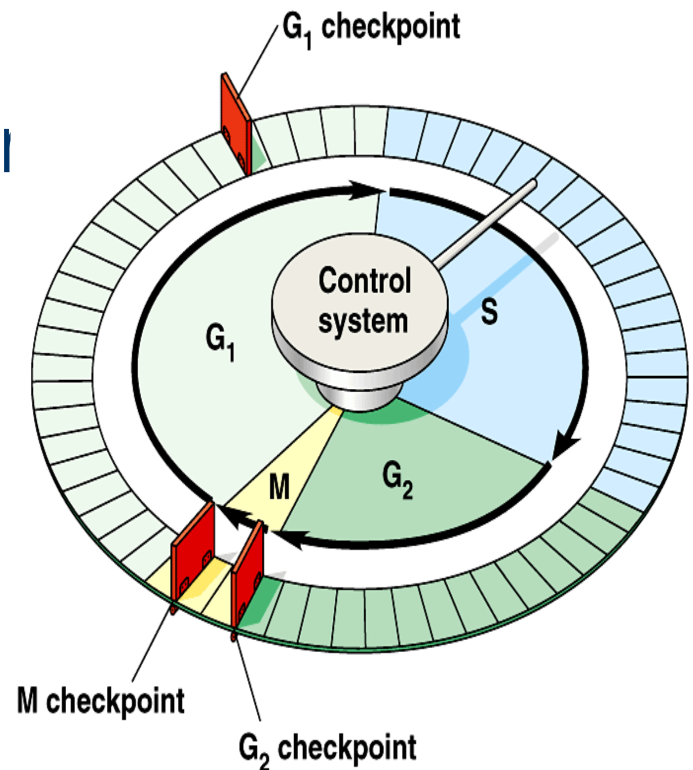
Overview of Cell Cycle Control

- Two irreversible points in cell cycle
  - *replication of genetic material
  - *separation of sister chromatids
- Checkpoints
  - process is assessed & possibly halted



Checkpoint control system

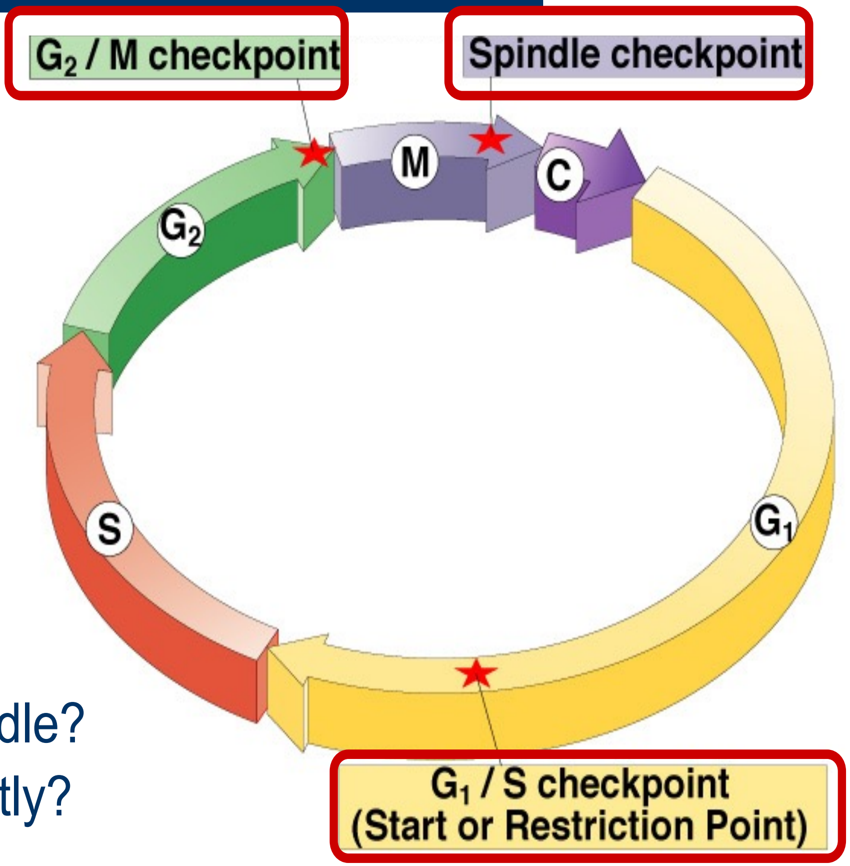
- Checkpoints
  - cell cycle controlled by STOP & GO chemical signals at critical points
  - signals indicate if key cellular processes have been completed correctly



Checkpoint control system

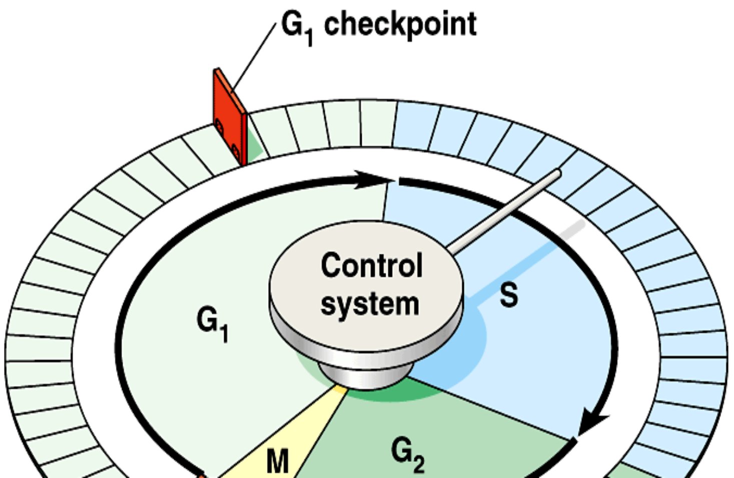
3 major checkpoints:

- *G_1/S
  - can DNA synthesis begin?
- *G_2/M
  - has DNA synthesis been completed correctly?
  - commitment to mitosis
- *spindle checkpoint
  - are all chromosomes attached to spindle?
  - can sister chromatids separate correctly?



G_1/S checkpoint

- G_1/S checkpoint is most critical
  - primary decision point
    - "restriction point"
  - if cell receives "GO" signal, it divides
    - internal signals: cell growth (size), cell nutrition
    - external signals: "growth factors"
  - if cell does not receive signal, it exits cycle & switches to G_0 phase
    - non-dividing, working state



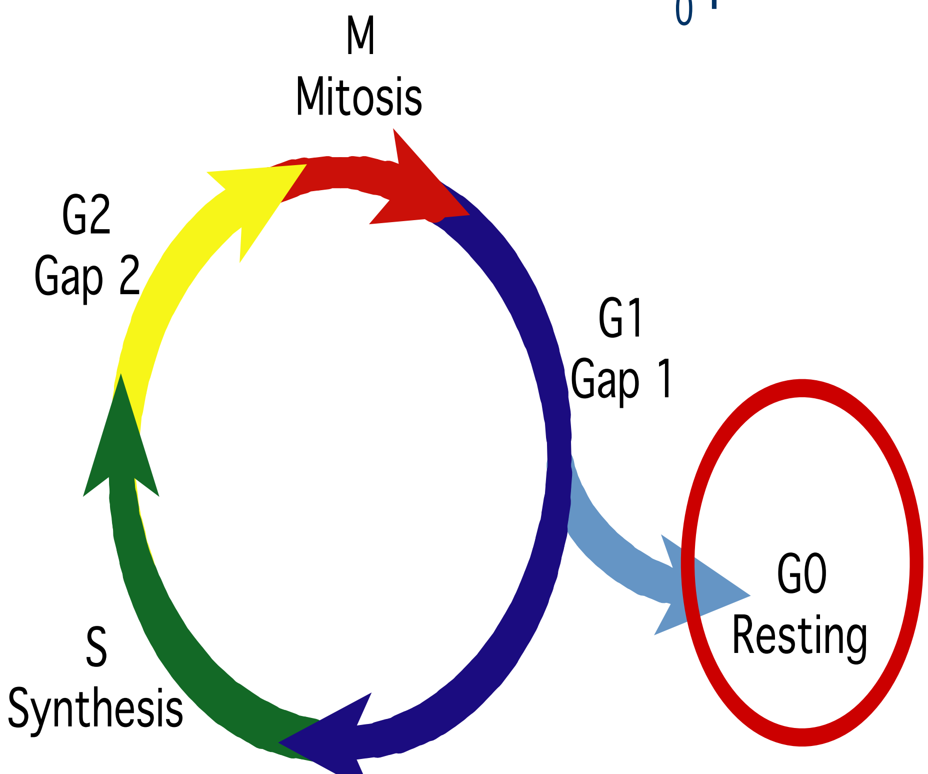
- ***The G1/S Checkpoint***
- The G1 checkpoint determines whether all conditions are favorable for cell division to proceed. The G1checkpoint, also called the restriction point, is the point at which the cell irreversibly commits to the cell-division process. In addition to adequate reserves and cell size, there is a check for damage to the genomic DNA at the G1 checkpoint. A cell that does not meet all the requirements will not be released into the S phase.

- ****The G2/M Checkpoint**
- The G2 checkpoint bars the entry to the mitotic phase if certain conditions are not met. As in the G1checkpoint, cell size and protein reserves are assessed. However, the most important role of the G2checkpoint is to ensure that all of the chromosomes have been replicated and that the replicated DNA is not damaged.

- *****The(spindle) M Checkpoint**
- The M checkpoint occurs near the end of the metaphase stage of mitosis. The M checkpoint is also known as the spindle checkpoint because it determines if all the sister chromatids are correctly attached to the spindle microtubules. Because the separation of the sister chromatids during anaphase is an irreversible step, the cycle will not proceed until the kinetochores of each pair of sister chromatids are firmly anchored to spindle fibers arising from opposite poles of the cell.

G_0 phase

- G_0 phase
  - non-dividing, differentiated state
  - most human cells in G_0 phase



liver cells

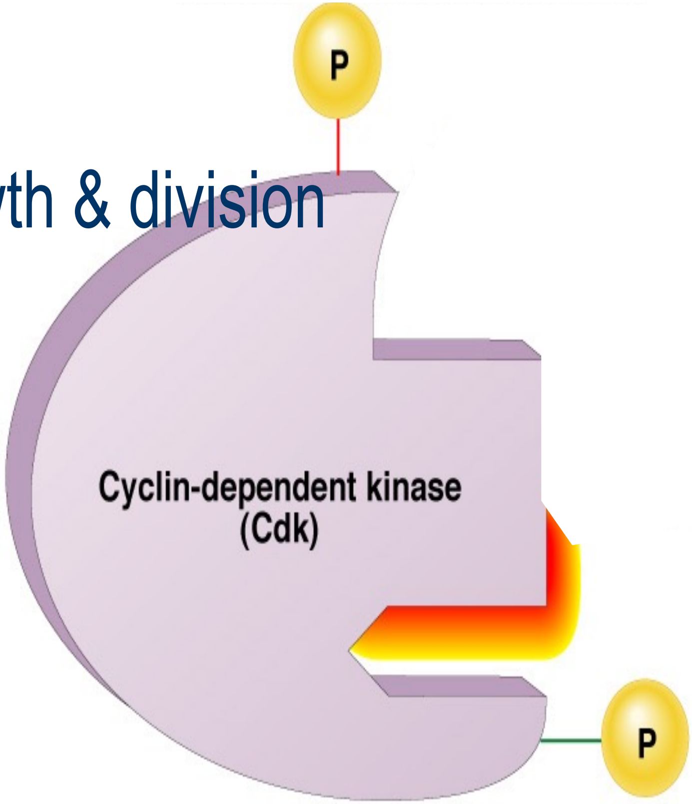
in G_0, but can be "called back" to cell cycle by external cues

nerve & muscle cells

highly specialized arrested in G_0 & can never divide

"Go-ahead" signals

- Protein signals that promote cell growth & division
  - internal signals
    - "promoting factors"
  - external signals
    - "growth factors"
- Primary mechanism of control
  - phosphorylation
    - kinase enzymes
    - either activates or inactivates cell signals



How Cell Cycle Control System Functions:

Basically through **cyclically** activating and inactivating (phosphorylation and dephosphorylation) of key proteins and protein complexes.

Four major types of proteins are involved :

1. Kinases Cdks (Phosphorylation) → core of the cell-cycle control system present in proliferating cells throughout the cell cycle ,activated / inactivated.
2. Phosphatases (Dephosphorylation)
3. Cyclins → Their concentrations vary in a cyclical fashion during the cell cycle
4. Cdks inhibitor proteins.
5. Anaphase promoting complex (APC)

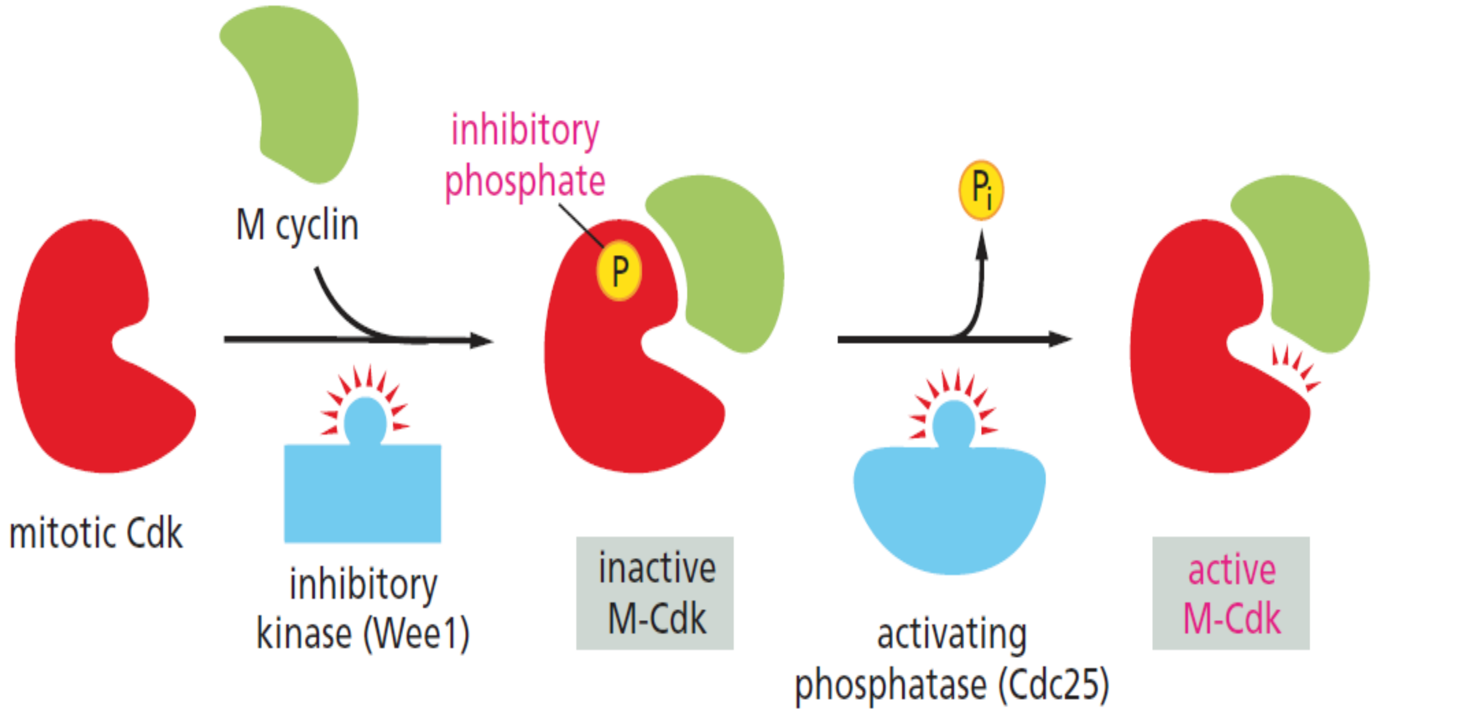
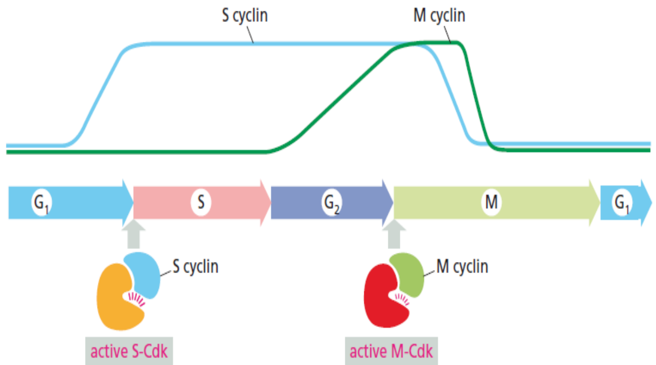


TABLE 18–2 THE MAJOR CYCLINS AND CDKS OF VERTEBRATES

Cyclin–Cdk Complex	Cyclin	Cdk Partner
G_1-Cdk	cyclin D*	Cdk4, Cdk6
G_1/S-Cdk	cyclin E	Cdk2
S-Cdk	cyclin A	Cdk2
M-Cdk	cyclin B	Cdk1

*There are three D cyclins in mammals (cyclins D1, D2, and D3).




Thank you!!!
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